

EXPERIMENT NO. 8:

Write a program to implement Linear Regression (LR) algorithm in Python

Aim:

To implement the Linear Regression (LR) algorithm in Python and predict output values using a suitable dataset.

Procedure:

1. Import the required Python libraries.
2. Load the Diabetes dataset.
3. Split the dataset into training and testing sets.
4. Create the Linear Regression model.
5. Train the model using the training dataset.
6. Predict the output values for the test dataset.
7. Calculate the coefficient of determination (R^2 Score).
8. Display the predicted values and model accuracy.

Program:

Linear Regression Algorithm

```
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
```

Load Diabetes Dataset

```
diabetes = load_diabetes()
```

```
X = diabetes.data
```

```
y = diabetes.target
```

Split dataset into training and testing sets

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.3, random_state=42
```

)

Create Linear Regression model

```
model = LinearRegression()
```

Train the model

```
model.fit(X_train, y_train)
```

Predict test data

```
y_pred = model.predict(X_test)
```

```
print("First 10 Predicted Values:")
```

```
print(y_pred[:10])
```

Calculate R² Score

```
score = r2_score(y_test, y_pred)
```

```
print("\nR2 Score:")
```

```
print(round(score, 2))
```

Output:

```
... First 10 Predicted Values:
[138.46970031 181.10052342 125.34400904 292.75977277 123.8830531
 91.89830434 257.26463123 177.76169318 84.98549706 109.15960992]

R2 Score:
0.48
```

Result:

The Linear Regression (LR) algorithm was successfully implemented in Python using the Diabetes dataset. The model predicted the output values and achieved an R² score of approximately 0.48, demonstrating the performance of the linear regression model.